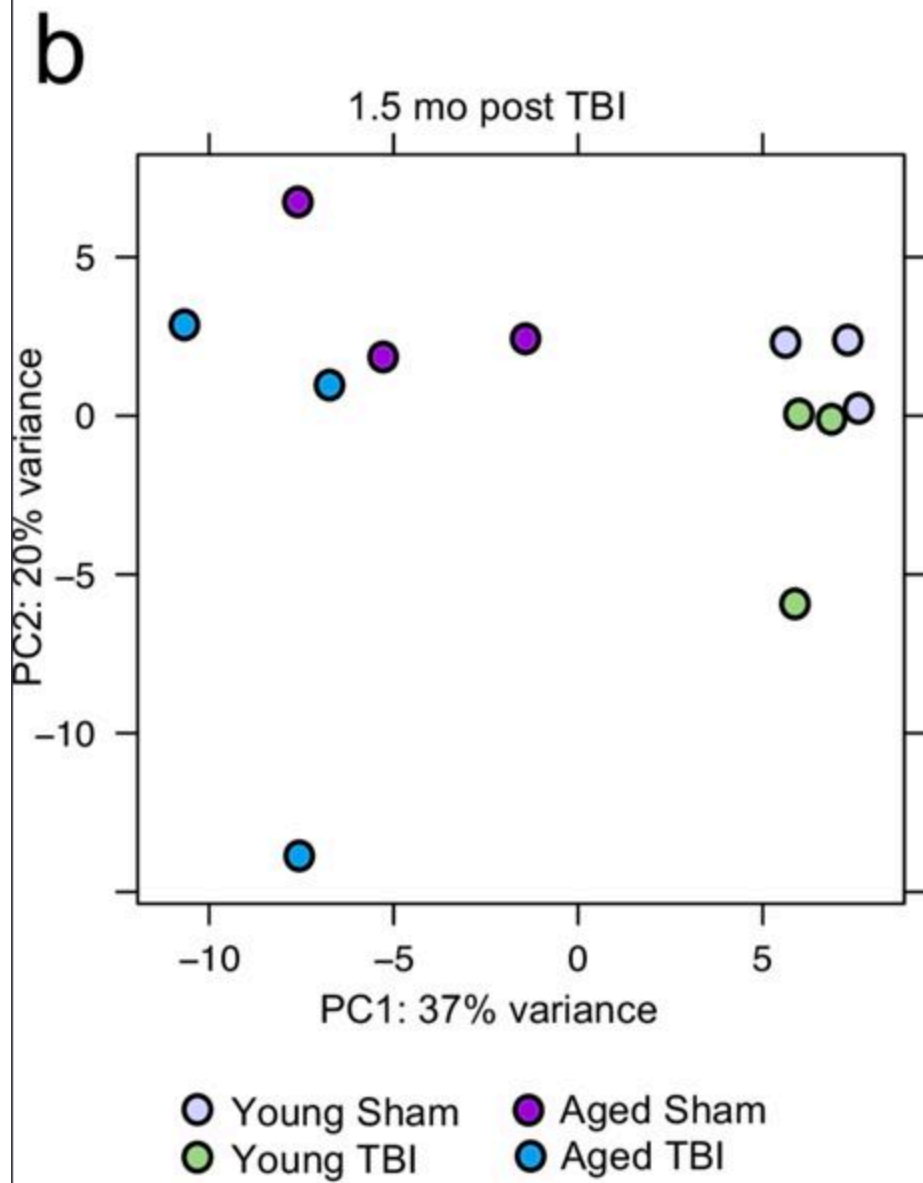


Milana Serkin

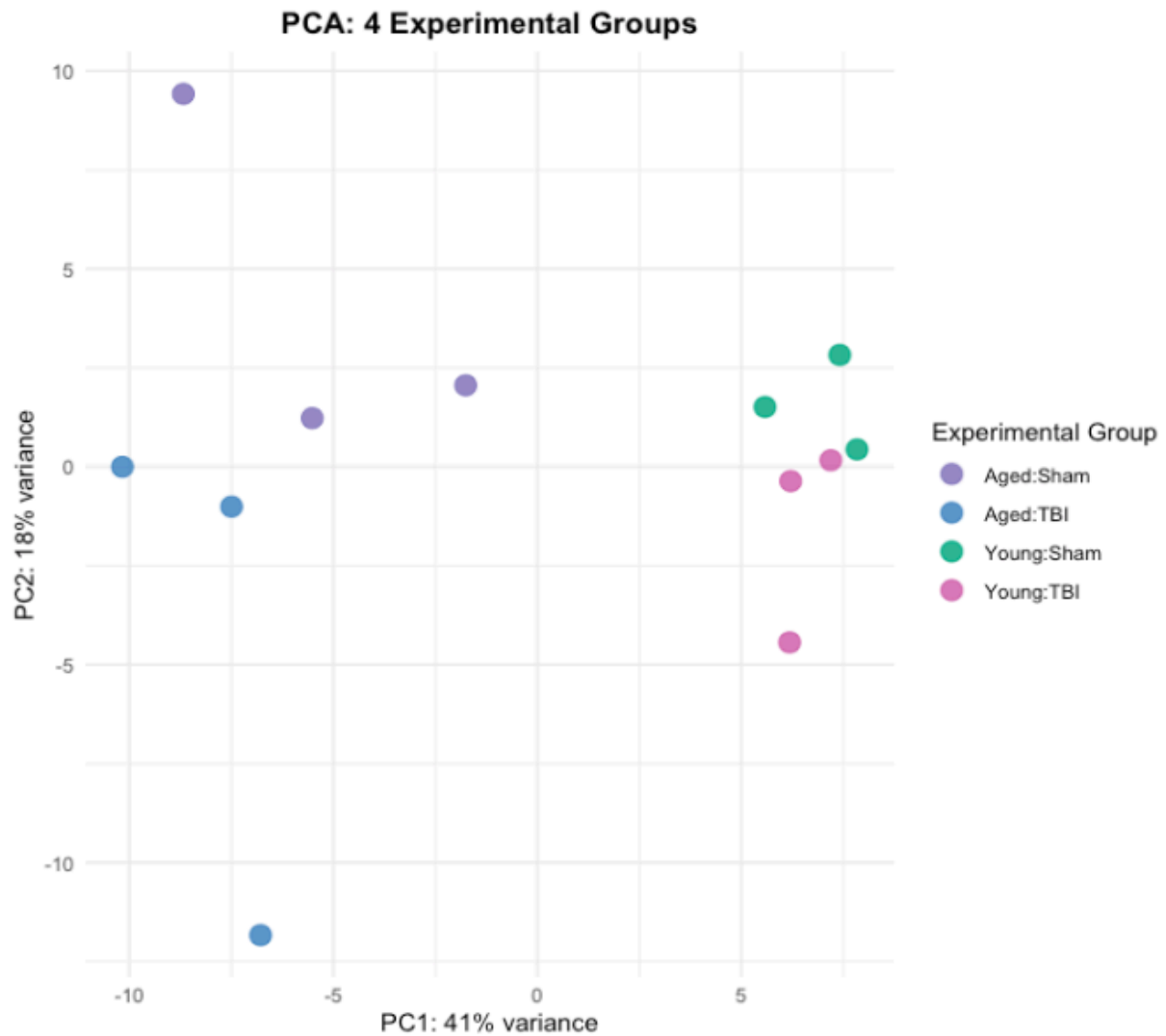
From my research paper I chose for the course: Ashley C Bolte, Daniel A Shapiro, Arun B Dutta, Wei Feng Ma, Katherine R Bruch, Michael A Kovacs, Ana Royo Marco, Hannah E Ennerfelt, John R Lukens (2023) The meningeal transcriptional response to traumatic brain injury and aging eLife 12:e81154.

This paper is about traumatic brain injury in mice and how age affects recovery. The authors used bulk RNA-sequencing on four groups of mice (Young Sham, Young TBI, Aged Sham, Aged TBI) to discover how the aging brain's transcriptomic response to injury differs from a healthy, young brain's response. I chose to take this small dataset and re-create the results from DESeq2, PCA, and volcano plots. I then furthered the analysis with Pathview.

PCA from paper:

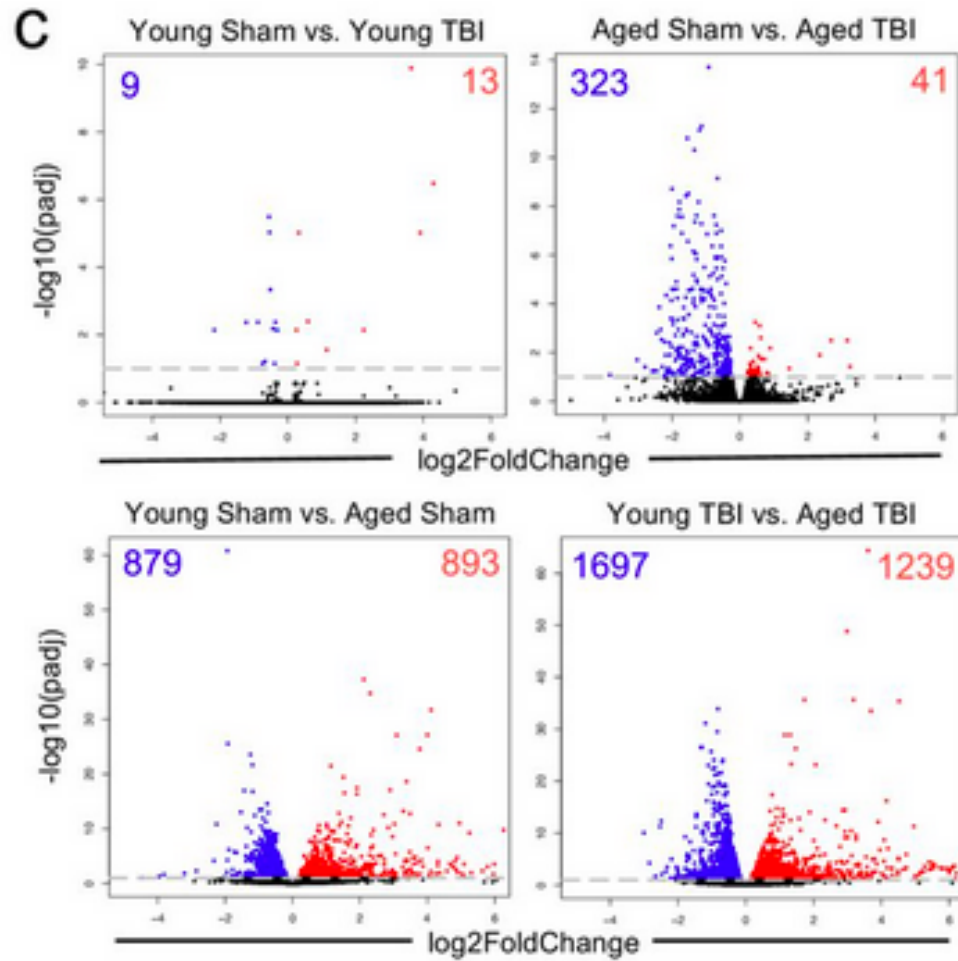


My PCA:

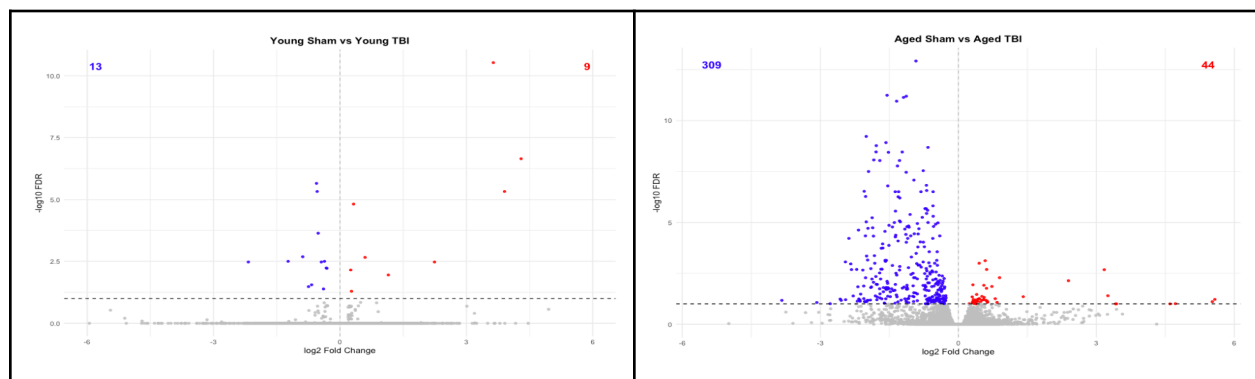


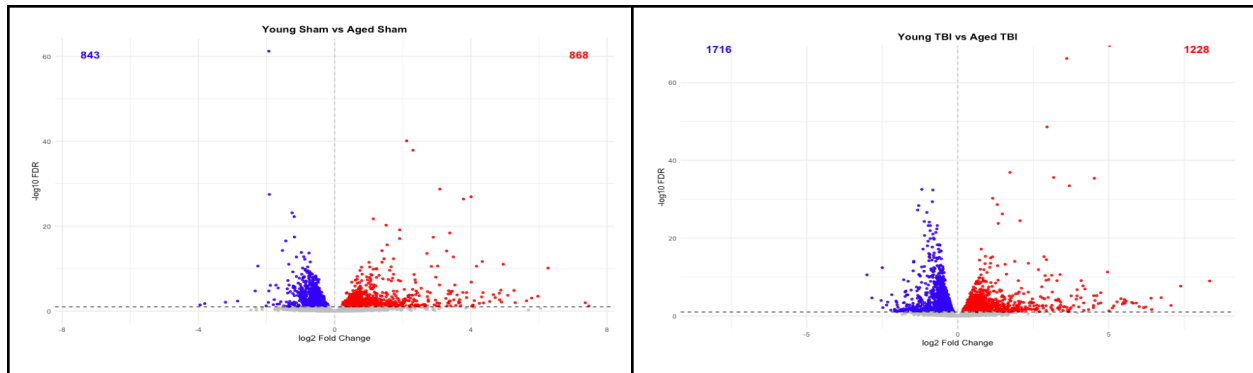
I was able to get similar results. The percent variance varies by under 4% however the same clustering patterns are present in both. I consider this a successful replication.

Their Volcano Plots:



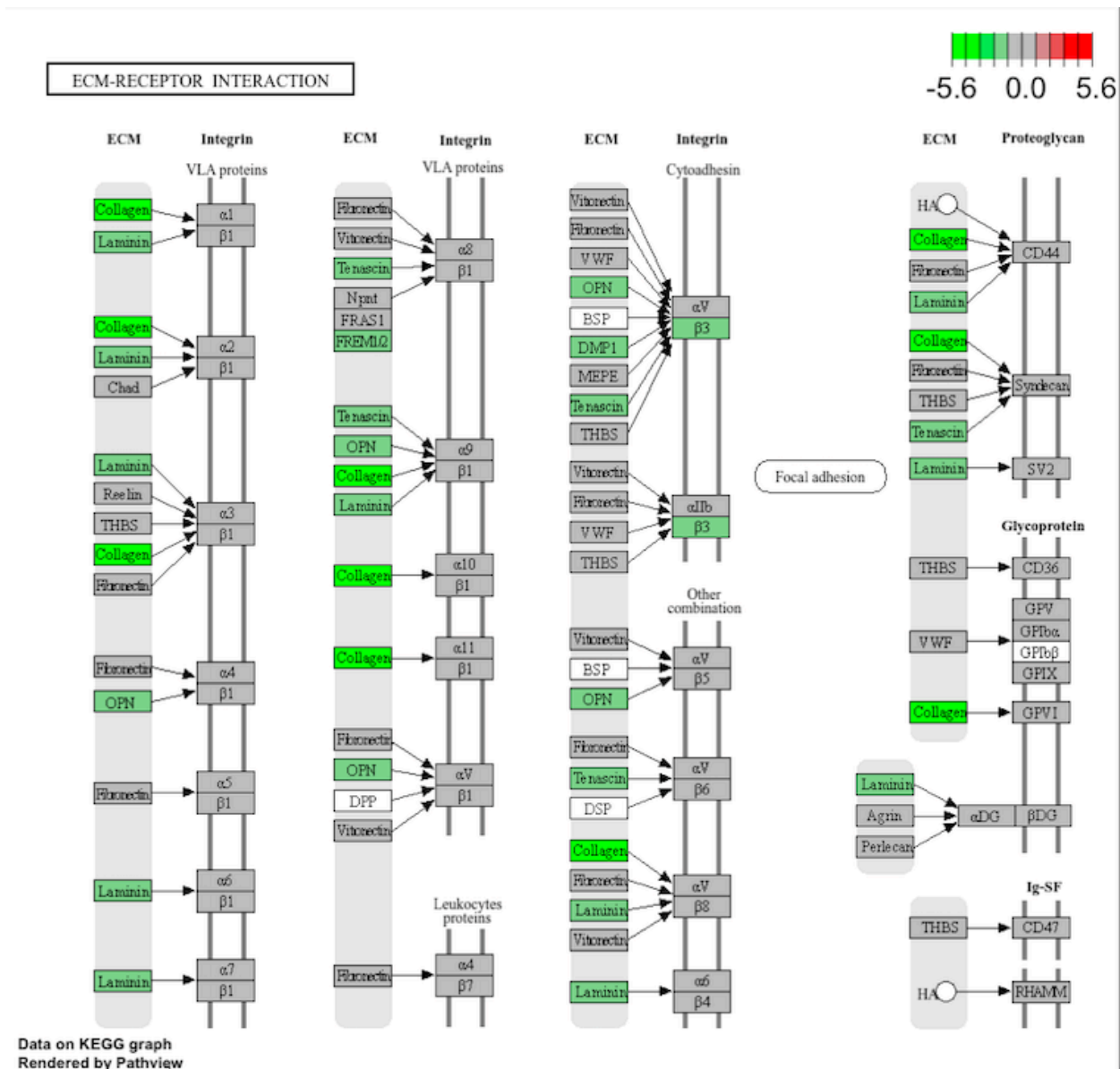
My volcano Plots:





Looking at the patterns of the volcano plots, the up and down expressed genes, the patterns are the same for each category. The exact numbers vary however that can be due to a variety of factors. The same patterns can be seen in both the papers and my plots. I consider this a successful replication.

Pathviewer in Aged mice with TBI:



To further my analysis, I ran pathviewer in R using gage and pathview to map significant Differentially Expressed Genes (DEGs) to the KEGG pathway database. The most significantly down regulated pathway was the Extracellular Matrix (ECM)-receptor interaction pathway (KEGG ID: mmu04512). The ECM plays a role in structural support and allowing cells to interact with other cells. The down regulated genes (green boxes) show a failure in the brain's ability to maintain structural integrity after an injury. This can explain why aged mice recover slower than young mice after a TBI.

The ECM pathway showed significant downregulation in aged TBI mice ($p = 0.016$). The pathway visualization reveals that key structural proteins were suppressed, including Collagen isoforms (Col1a, Col2a, Col10a, Col11a), Laminin (Lama1, Lama2, Lama5), Tenascin, Osteopontin (OPN), and Thrombospondin (THBS). These proteins help form the structural of the

brain's extracellular matrix and therefore are essential for maintaining the blood/brain barrier and supporting neuronal recovery. Their coordinated downregulation suggests that aged brains fail to maintain structural integrity following TBI, likely explaining the slower recovery.

This aligns with the findings in the paper, as they found that aged TBI mice exhibited downregulation of genes involved in extracellular matrix remodeling, unlike young mice where the meningeal transcriptome largely returned to its pre-injury state. The paper also mentioned that one week following injury in young mice, fibroblasts exhibited 368 activated genes including those important for collagen production and extracellular matrix remodeling. This activation failed to persist for aged mice.